

Fardeen Bablu

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EDUCATION

Vanderbilt University — Nashville, TN

May 2026

Bachelor of Science in Computer Science, Mathematics, Public Health

Activities: Head Resident Advisor, Entrepreneurship Society, First Generation VU, Assoc. of Bengali Students

SKILLS

Programming: Python, JavaScript/TypeScript, Swift, Rust, SQL

Libraries: PyTorch, React, React Native, Expo, Svelte, Node.js, Firebase, Tauri, Three.js

Tools: Figma, Git, Docker, Stripe, Plaid, AWS, GCP, CI/CD

EXPERIENCE

Teaching Assistant

Nashville, TN

Vanderbilt University, Department of Computer Science

Jan. 2025 – Feb. 2026

- Held 12 hours/week of office hours for 50+ students on Java fundamentals and general debugging, OOP, & algorithm design.
- Graded 10+ coding assignments per cycle via GitHub & Autograder; improved student resubmissions by 30%.
- Supported 8 Software Engineering course teams; advised on design patterns, testing, and full-stack development.

Software Engineering Intern

Washington, DC

Jenner & Block LLP

June 2024 – Aug. 2024

- Led digital transformation initiative for 50+ attorneys, automating workflows to reduce onboarding costs by 40%.
- Configured Cisco Meraki Access Points across office; improved wireless stability & cut IT tickets by 25%.
- Collaborated with cross-functional teams to integrate 5+ enterprise applications, streamlining operations.

PROJECTS

Niyah - Focus Commitment App - React Native, Swift, Firebase, Stripe, Plaid - niyah.live

Jan. 2026 – Present

- Shipped commitment-stake focus app; Swift modules on Screen Time API earned FamilyControls Distribution.
- Engineered Stripe Connect + Plaid integration across 33 Firebase Cloud Functions for live payments.
- Led React Native + Expo iOS build to TestFlight with 1,018 tests, EAS pipeline, and Sentry observability.

Vectr - Pre-Scene Intelligence for EMTs - Svelte 5, Rust, Tauri, PostGIS - [Whitepaper](#)

Jan. 2026 – Present

- Trained DINOv2 on 30K satellite tiles for helicopter LZ detection; deployed county-wide with zero FAA helipads.
- Built mobile AI scribe converting on-scene audio into ePCR fields, cutting 30 min/shift of charting.
- Piloting with Greene County EMS on 2 units after 3-day ride-along; presented to directors in TN and VA.

Sentinel - LLM Safety Research Dashboard - React, Express, MongoDB, Python, Ollama

Feb. 2026 – Present

- Designed context-window dashboard at Vanderbilt's Wicked Problems Lab; refusal decay across 5 fill levels.
- Ran 300 adversarial probes on HarmBench/AdvBench/JailbreakBench across GPT-4o, Claude, DeepSeek-R1.
- Wired React + Recharts frontend to Express + MongoDB API with power-law fits and Wilson CIs.

Spevents.live - Real-time Photo Wall - React, Three.js, Tailwind CSS, AWS S3 - [Live](#)

Dec. 2025

- Launched serverless platform with React Three Fiber and S3 + CloudFront, serving 200+ photos at 4 events.
- Created REST API on Vercel with Firebase Auth/Firestore and QR-based zero-download uploads.
- Designed 5 displays (Grid, Slideshow, 3D, Parallax) with Framer Motion and Hugging Face moderation.

QuizLITE - Flashcard Desktop App - C++17, SQLite, Qt6, CMake, GTest - [Github](#)

Sept. 2024

- Built flashcard app with 3 adaptive modes prioritizing low-accuracy questions via spaced repetition.
- Architected SQLite backend with singleton-pattern CRUD operations and per-question accuracy tracking.
- Configured GitHub Actions with 35+ GoogleTest cases across Ubuntu/macOS, clang-format linting.

AWARDS

Amazon Mission Autonomy Hackathon 2025 - 2nd Place (\$15K Prize) - [Pitch](#) | [Github](#)

Nashville, TN

HUSH-MESH, an AWS-powered autonomous maritime defense network with RF-silent drones for routing.

Vanderbilt X Ironsite Hackathon 2025 - 3rd Place (\$2.5K Prize) - [Pitch](#) | [Github](#)

Nashville, TN

Built an object permanence model to track blocked objects, boosting accuracy from 12% to 94%.

Google Solutions Challenge 2025 - 1st Place - [Pitch](#) | [Github](#)

New York, NY

Gemicast, a real-time outage-risk dashboard + predictive alerts, inspired by recent Tennessee floods.